Detection and Biomass Estimation of *Phaeocystis globosa* Blooms off Southern China From UAV-Based Hyperspectral Measurements

Xue Li, Shaoling Shang, Zhongping Lee, *Member, IEEE*, Gong Lin, Yongnian Zhang, Jingyu Wu, Zhenjun Kang, Xiangxu Liu, Cheng Yin, and Yue Gao

***Abstract*— *Phaeocystis globosa* (*P. globosa*) is a unique causative species of harmful algal blooms, which can form gelatinous colonies. We, for the first time, used unmanned aerial vehicle (UAV) measurements to identify *P. globosa* blooms and to quantify the biomass. Based on *in situ* measured remote sensing reflectance (R_{rs}), it is found that, for *P. globosa* blooms, the maximum of the second-derivative ($d\lambda^2 R_{rs}$) of $R_{rs}(\lambda)$ in the 460–480-nm domain is beyond 466 nm. An analysis of the absorption properties from algal cultures suggested that this feature comes from the absorption of chlorophyll c_3 (Chl−/c_3) around 466 nm, a prominent feature of *P. globosa*. This position of $d\lambda^2 R_{rs}$ maximum was, thus, selected as the criterion for *P. globosa* identification. The spatial extent of *P. globosa* blooms in two bays off southern China was then mapped by applying the criterion to UAV-measured R_{rs}. Twelve out of 16 UAV and *in situ* match-up stations were consistently identified as dominated by *P. globosa*, indicating the accuracy of 75%. Furthermore, using localized empirical models, chlorophyll a (Chl−/*a*) concentration and colony numbers of *P. globosa* were estimated from UAV-derived R_{rs}, where *P. globosa* colonies were found in a range of ∼3–37 gel matrix/L, indicating the occurrence of weak to moderate *P. globosa* blooms during the surveys. The promising results suggest a high potential for detection and quantification of *P. globosa* blooms in near-shore bays or harbors using UAV-based hyperspectral remote sensing, where conventional ocean color satellite remote sensing runs into difficulties.**

***Index Terms*— Diatoms, harmful algal blooms (HABs), hyperspectral, *Phaeocystis globosa* (*P. globosa*), quantification, unmanned aerial vehicle (UAV).**

I. Introduction

In A dynamic era of accelerated climatic and environmental changes, it is apparent that harmful algal blooms (HABs) have become more frequent, intensive, and widespread, causing increasing economic losses in coastal waters [1]–[3]. Although HABs are a worldwide problem, they appear particularly severe in China, where a dramatic increase in the frequency and diversity of HABs, and changes in the composition of HABs species have occurred, partially due to environmental stresses arising due to the country's rapid economic development [4]. A severe HAB event of *Phaeocystis globosa* (*P. globosa*) (Prymnesiophyceae), which is a high-biomass bloom-forming species, was first reported in the literature in 1997 in coastal waters of Guangdong and Fujian provinces, China [5], [6]. After that earliest record, *P. globosa* became a recurrent bloom species, with 70 outbreaks documented from 1997 to 2019 in China seas [7]. These blooms pose considerable threats to water quality and ecosystem health, as they may form gelatinous colonies of up to 3 cm in diameter during blooms [8]–[10], leading to accumulation of odorous foams and release of toxins, thus negatively affecting aquaculture, tourism, and even the supply of cooling water for nuclear power plants [11], [12]. In addition, *P. globosa* may produce dimethylsulfide, playing a fundamental role in the global biogeochemical cycle of carbon and sulfur, thus leading to concern by the climate change community [13], [14]. It is noteworthy that *P. globosa* blooms are a global event, very common in European coastal waters [15]–[19], and also found in coastal waters off Vietnam [20]. Monitoring *P. globosa* blooms, for example, using satellite- or aircraft-based remote sensing measurements, is, therefore, critically needed, as a rapid and reliable method is demanded to determine the magnitude and extent of such blooms.

Indeed, such an effort was initiated more than a decade ago, as *P. globosa* blooms occur after diatom blooms almost every spring in European coastal waters [18], [21]–[23], where *in situ* optical measurements of *P. globosa* and diatom blooms indicated different optical signals between them [24]–[27].

Manuscript received August 14, 2020; revised November 25, 2020 and January 7, 2021; accepted January 8, 2021. Date of publication January 26, 2021; date of current version December 3, 2021. This work was supported in part by the Chinese Ministry of Science and Technology through the National Key Research and Development Program of China under Grant #2018YFC1407504, Grant #2017YFC1404801, Grant #2016YFE0202100, and Grant #2016YFA0601201; and in part by the National Natural Science Foundation of China under Grant #41576169 and Grant #41830102. *(Corresponding author: Shaoling Shang.)*

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Digital Object Identifier 10.1109/TGRS.2021.3051466

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Lubac *et al.* [28] proposed multispectral and hyperspectral approaches to detect *P. globosa* blooms based on field measurements in the eastern English Channel and the southern North Sea. The latter approach, which took advantage of the second derivative ($d\lambda^2 R_{rs}$) of a remote sensing reflectance (R_{rs}, $sr^{-/1}$) spectrum, was easy to perform and yielded encouraging results. Astoreca *et al.* [25] further developed an algorithm to retrieve Chl-/c_3 in waters dominated by *P. globosa* using either absorption coefficients or water-leaving reflectance. Positive results were obtained when *P. globosa* cell density exceeded 3×10^6 cells/L. Extending the method presented in Miller *et al.* [29], Kurekin *et al.* [30] developed a multivariant classifier to flag ocean color pixels as either "harmful bloom," "nonharmful bloom," "no bloom," or "unknown," where the inputs for such a classifier include water-leaving radiance and its spectral ratios, absorption coefficient, and its spectral ratios. Two separate classifier systems were developed: one for *Karenia mikimotoi* bloom and the other for *P. globosa* bloom. However, the application of this system to an ocean color image requires *a priori* knowledge of bloom species and assumes it is the same type of phytoplankton species for the wide areas covered in an ocean color image. In other words, it does not distinguish *Karenia mikimotoi* from *P. globosa* from ocean color data. To date, there has been no report showing successful detection and quantification of *P. globosa* blooms from satellite observations.

The reason for this shortage is, in part, due to that very rare hyperspectral and high-spatial-resolution sensors onboard satellites are available. One successful report is an evaluation of a ciliate bloom's distribution at a spatial resolution of 100 m in western Long Island Sound using hyperspectral imagery data collected onboard the International Space Station [31]. On the other hand, *P. globosa* blooms occur not only usually in coastal regions or inshore bays but also patchy or with a small spatial coverage and fast-changing. It is difficult to have reliable measurements of these blooms with satellite sensors of coarse spatial resolution and low revisit time, not to mention missing data due to the blocking effects of clouds, especially when timely monitoring of bloom location and extent is required. With the rapid development of technologies associated with unmanned aerial vehicles (UAVs), measurement via UAV is an important and viable way to fill the gap, as it can map the bloom details with very high spatial resolution from a low flying altitude, even under heavy clouds.

Here, in this study, using data from *in situ* and UAV surveys of *P. globosa* and diatom blooms in coastal waters of southern China, a hyperspectral approach for identification of *P. globosa* blooms developed by Lubac *et al.* [28] was tested. Furthermore, the biomass of *P. globosa* was quantified from UAV acquired R_{rs}. The overarching objective is to assess the feasibility of identifying and quantifying *P. globosa* blooms using UAV-based hyperspectral measurements.

II. Data and Measurements

A. In Situ Measurements

In situ measurements were carried out in Xiamen Bay (XMB, June 17–18, 2015), where the diatom *Chaetoceros curvisetus* bloomed, and in Weitou Bay (WTB, September 11–12, 2015) and Qinzhou Bay (QZB, January 23–26, 2019), where *P. globosa* bloomed (see bay locations and sampling stations in Fig. 1). It is necessary to keep in mind that: 1) normally, the dominant phytoplankton species is diatom in WTB and QZB [32]–[35], while *P. globosa* almost blooms each winter in QZB [35] and 2) *P. globosa* blooms often occur after diatom blooms [18], [21]–[23]. Therefore, blooms of other phytoplankton species are excluded in this study, with a focus on the differentiation of *P. globosa* blooms from diatom blooms using UAV-based hyperspectral measurements.

Major phytoplankton species were identified microscopically from collected surface samples, without cell enumeration. Concentrations of chlorophyll a (Chl-/*a*, mg/m^3) were measured from water samples fluorometrically [36]; they were found ranging 33.1–133.1 mg/m^3 in XMB, 7.3-45.6 mg/m^3 in WTB, and 2.4–12.7 mg/m^3 in QZB, respectively.

Remote sensing reflectance is the ratio of water-leaving radiance (L_w) to downwelling irradiance just above the surface [$E_d(0+)$] [37]. During these field surveys, L_w and $E_d(0+)$ were measured *in situ* using a GER1500 spectroradiometer (Spectra Vista Corporation, Poughkeepsie, NY, USA), which covers a spectral range of 350–1050 nm with a resolution of 3 nm. The parameter R_{rs} was further obtained following the method detailed by Shang *et al.* [38]. Briefly, after total reflectance (T_{rs}, $sr^{-/1}$, the ratio of total upwelling radiance to downwelling irradiance) and sky reflectance (S_{rs}, $sr^{-/1}$, the ratio of sky radiance to downwelling irradiance) are calculated from GER1500 measurements, R_{rs} can be obtained by deducting the reflected S_{rs} from T_{rs} and further removal of the residual surface contribution (δ) through iterative derivations described by Lee *et al.* [39].

B. UAV Measurements and Data Processing

UAV overflights were carried out for the detection of *P. globosa* blooms in WTB on September 12, 2015, and QZB on January 25, 2019, within 9:00–16:30 local time [see flight tracks in Fig. 1(b) and (d)]. As described by Shang *et al.* [40], the UAV used in the surveys was an LT-150 (TOPRS Technology Company Ltd., China) equipped with an AvaSpec-dual spectroradiometer (Avantes, The Netherlands). Two sensors are factory calibrated and cover a spectral range of 360–1000 nm with a resolution of ~1 nm and a signal-to-noise ratio of ~100–400 in the visible domain. Two properties were measured separately with the two sensors: total upwelling radiance ($L_t(h, \lambda, \text{and } t)$, $W/m^2/nm/sr$) and downwelling irradiance at UAV altitude ($E_d(h, \lambda, \text{and } t)$, $W/m^2/nm/sr$), where h is the flight height of the UAV (300 m herein) and t is the time when the measurements were taken. The integration time for the two sensors could be adjusted in a range of 2 μs–20 s. An optimized setting of 6 ms for the L_t sensor and 3 ms for the E_d sensor was determined according to our previous experiments [40] although more experiments under various weather and water conditions are certainly needed to explore if any adjustments of integration time may help to improve data quality. The resulting size of

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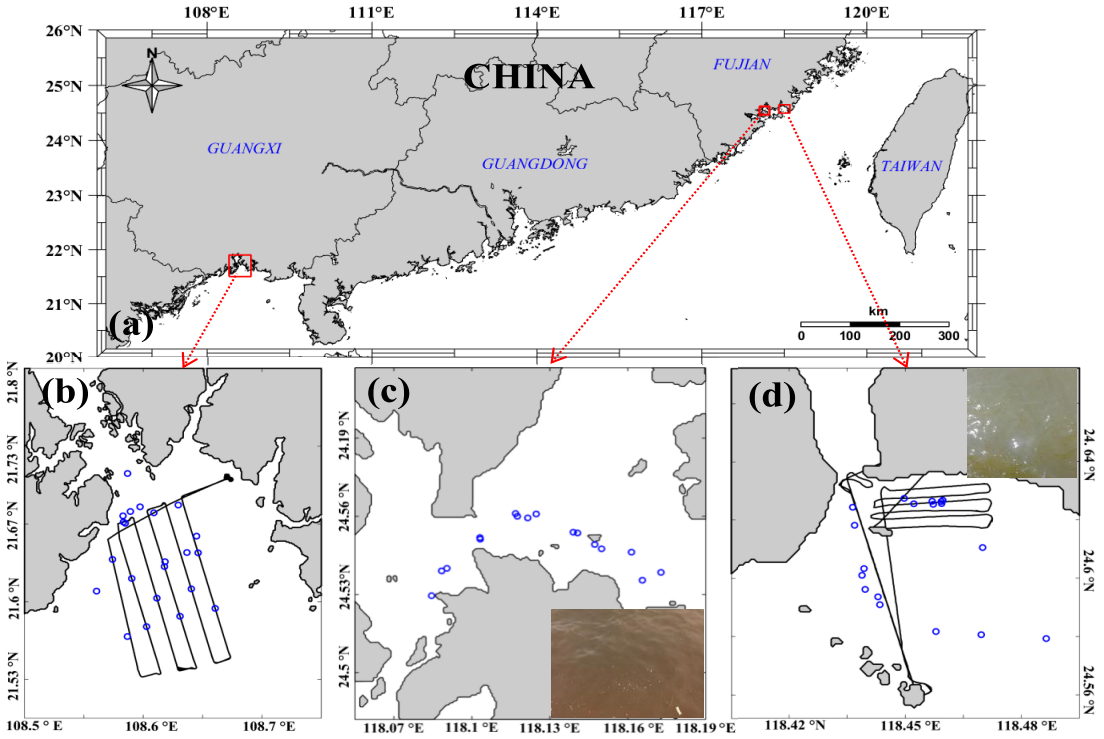


Fig. 1. (a) Map of the seas off southern China, showing study locations; *in situ* surveys (blue circles) with UAV overflights (black lines) in (b) QZB, (c) XMB, and (d) WTB; the two inset photographs in (c) and (d) show the coloration of the water during surveys taken under intensive bloom conditions in XMB and WTB, respectively.

the circular footprint (hereafter for simplicity, termed spot) on the grounds from the spectroradiometer for L_t is ~5.2 m in diameter, and the distance between two consecutive spots is ~30 m. Furthermore, for simplicity, h, λ, and t will be omitted hereafter unless it is necessary.

Processing to obtain R_{rs} from UAV measurements followed the procedure of Shang *et al.* [40]: 1) a quality control (QC) procedure, including wavelet transform and sun glint detection, was implemented to exclude outliers generated due to tilts of the UAV platforms and moving clouds and those with severe sun glint affecting the raw data, i.e., L_t and E_d; 2) T_{rs} was obtained as the ratio of L_t to E_d; and 3) R_{rs} was calculated by deducting the reflected S_{rs} and residual surface contribution from T_{rs} through iterative derivations following Lee *et al.* [39], with S_{rs} at 500 nm included in the optimization procedure.

A total of 1826 and 2993 paired L_t and E_d data were collected from UAV measurements in the WTB and the QZB, respectively. After the first processing step mentioned above [annotated as QC in Fig. 2(a)], the numbers became 716 and 1603 in the WTB and the QZB, respectively [see Fig. 2(a)]. After steps #2 and #3, the resulting R_{rs} was further assessed using a novel quality assurance system [41] (termed QA), where each R_{rs} spectrum will be measured with a QA score. To ensure high-quality data, only QA scores > 0.75 were retained for use in subsequent analysis. The resulting R_{rs} numbers were 388 (~21% of raw data) and 881 (~29% of raw data) in the WTB and the QZB, respectively [see Fig. 2(a)]. These measures of data numbers indicate that more flights are necessary if a significantly larger number of high-quality R_{rs} spectra are required with today's UAVs technology.

In [40], for rigorous method evaluation, a spatiotemporal window of 500 m and 30 min was used to collect match-ups of R_{rs} between UAV and *in situ* measurements. Here, with a focus on bloom detection and in view of that these blooms do not dissipate in hours, the time window of the matchup is set as the same day, while the spatial window as 500 m, which resulted in 16 match-ups, with nine in WTB and seven in QZB. Representative R_{rs} spectra are shown in Fig. 2(b) and (c) as a demonstration, indicating quite consistent spectra between UAV-derived and *in situ* measured R_{rs}. However, the deviation in the blue bands (~400-440 nm) is also evident, particularly in QZB, probably associated with the removal of surface reflected sky light, which contributes more to the blue wavelengths. Some deviations also appear in the range of 625–630 nm. Root-mean-square difference (RMSD) and averaged unbiased absolute percentage difference ($\boldsymbol{\varepsilon}$) were calculated to assess the consistency between field-measured ($R_{rs}{}^F$) and UAV-derived R_{rs} ($R_{rs}{}^U$) for the spectral range of 400–700 nm

$$\mathrm{RMSD} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}\left(R_{rs}{}^F - R_{rs}{}^U\right)^2} \tag{1}$$

$$\boldsymbol{\varepsilon} = \left[\frac{1}{n}\sum_{i=1}^{n}\frac{\left|R_{rs_i}{}^F - R_{rs_i}{}^U\right|}{R_{rs_i}{}^F + R_{rs_i}{}^U}\right] \times 200\%. \tag{2}$$

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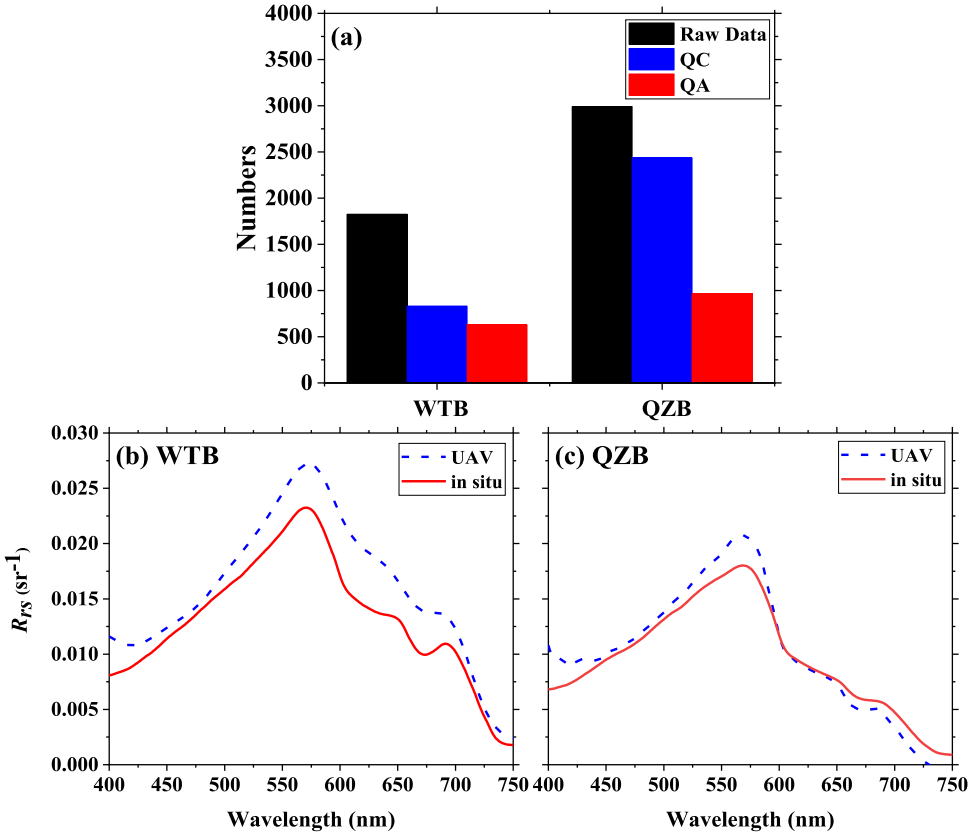


Fig. 2. (a) Number of remote sensing reflectance (R_{rs}) data obtained from UAVs in WTB and QZB. (b) and (c) Selected UAV-measured R_{rs} (blue) compared to concurrent and co-located *in situ* R_{rs} (red) in the two bays for demonstration, with a time window of <2 h and a spatial window of <500 m (see Section II-B for the explanation of QA and QC).

The resulted RMSD and ε ranged ~0.0013–0.0049 sr$^{-/1}$ and ~8%–47%, which are comparable to those previously reported for Fujian coastal waters, including WTB with a different temporal window (see [40]) and those reported in the literature when evaluating satellite R_{rs} in coastal waters [42]. Detailed information on these comparisons is listed in Table I.

C. Algal Cultures

Since we are short of *in situ* measurements of inherent optical properties and pigment composition of the bloom waters in XMB, WTB, and QZB, cultures of *P. globosa* and diatoms were grown in the laboratory to facilitate the characterization and differentiation of *P. globosa* from diatoms (see Section IV-A). Pure cultures of *P. globosa* and *Skeletonema costatum* (diatom) strains were obtained from the Center for Collections of Marine Algae, Xiamen University (http://mel2.xmu.edu.cn/gomap/ccma/index.asp). The *P. globosa* strain CCMA-166 was isolated from Beibu Gulf, where QZB is located, and the *Skeletonema costatum* strain CCMA-277 was isolated from XMB. Cultures were grown at 20 °C ± 2 °C under an irradiance of 2400 ± 200 lux with a 14 h:10 h light-dark photoperiod. The medium used for cultures is F/2 medium [43], prepared with 0.22-μm filtered seawater in sterilized conical flasks, and Si is added for the culture of *S. costatum*. Before measurements, the cultures were diluted to generate 1-L samples each of *P. globosa* and *S. costatum*, with Chl−/*a* concentrations ranging from 0.8 to 51.9 mg/m^3 and 0.5 to 92.0 mg/m^3, respectively. Measurements of absorption properties and pigment concentrations were performed on healthy cultures in exponential growth; the samples were assumed to be free of detritus. More specifically, as detailed by Shang *et al.* [44], the absorption coefficients were obtained with the filter-pad technique and were measured with a dual-beam PerkinElmer (PE) Lambda 950 spectrophotometer equipped with an integrating sphere (150 mm in diameter). Measurements of pigment concentrations were detailed in Wang *et al.* [45] and used a high-pressure liquid chromatography (HPLC) system (Agilent series 1100, Santa Clara, CA, USA) equipped with a 3.5-μm Eclipse XDB C^8 column (100 × 4.6 mm; Agilent Technologies).

D. Derivative Analysis of R_{rs}

Following Lubac *et al.* [28], the second derivative of R_{rs} ($d\lambda^2 R_{rs}$) for both *in situ* and UAV-derived R_{rs} at band i was calculated as

$$\left.\frac{d^2 R_{rs}}{d\lambda^2}\right| i \cong \frac{R_{rs}(\lambda_{i+1}) - 2R_{rs}(\lambda_i) + R_{rs}(\lambda_{i-1})}{(\Delta\lambda)^2} \tag{3}$$

where $\Delta\lambda$ is the band resolution, which is chosen to be 2.5 nm. This allowed a close examination of the peaks and troughs in R_{rs} and, thus, would facilitate the discrimination of *P. globosa* from diatoms. Before the computation of the derivative, data

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TABLE I
DETAILED EVALUATION INFORMATION OF THE 16 MATCHUP STATIONS

	RMSD* (sr^{-1})	$\varepsilon^{\#}$ (%)	*P. globosa* identification from UAV R_{rs}	*P. globosa* identification from in situ R_{rs}	Modeled Chl-*a* from UAV R_{rs} (mg/m^3)	in situ Chl-*a* (mg/m^3)	Modeled biomass from UAV R_{rs} (gel matrix/L)
WTB (2015.09.12)	0.0025	25	*P. globosa*	*P. globosa*	31.6	30.2	27
	0.0028	30	*P. globosa*	*P. globosa*	23.1	20.2	22
	0.0037	32	*P. globosa*	*P. globosa*	23.1	31.3	22
	0.0037	27	*P. globosa*	*P. globosa*	6.1	11.4	8
	0.0026	8	**diatom**	***P. globosa***	6.4	15.4	-
	0.0049	21	**diatom**	***P. globosa***	6.3	18.2	-
	0.0013	25	*P. globosa*	*P. globosa*	5.5	21.3	8
	0.0015	18	*P. globosa*	*P. globosa*	6.2	14.8	9
	0.0030	47	**diatom**	***P. globosa***	5.7	16.6	-
QZB (2019.01.25)	0.0017	21	*P. globosa*	*P. globosa*	2.4	4.4	4
	0.0014	11	*P. globosa*	*P. globosa*	4.1	5.6	6
	0.0030	26	*P. globosa*	*P. globosa*	10.0	7.4	12
	0.0035	25	*P. globosa*	*P. globosa*	5.9	7.4	8
	0.0027	27	**diatom**	***P. globosa***	7.2	5.7	-
	0.0030	42	*P. globosa*	*P. globosa*	5.6	12.7	8
	0.0033	38	*P. globosa*	*P. globosa*	8.6	7.0	11

* : Root-Mean-Square-Difference of R_{rs};
#: Average unbiased absolute relative difference of R_{rs} for wavelength in the 400-700 nm range.

smoothing was carried out using the Savitzky–Golay filter with a fourth-degree polynomial and a frame size of 21 [46].

III. DETECTION OF *P. globosa* BLOOMS FROM UAV HYPERSPECTRAL MEASUREMENTS

A. Approach to Identify P. globosa From R_{rs}

Fig. 3 shows *in situ* measured R_{rs} spectra of *P. globosa* and diatom blooms in WTB and XMB, respectively. Based on microscopic analysis and color changes of the diatom bloom water and the presence of visible gelatinous colonies in the *P. globosa* bloom water [see micrographs in Fig. 1(c) and (d)], it is certain that the waters were dominated by the two respective species. It is apparent that the R_{rs} spectra associated with the two blooms exhibited clear differences in both magnitude and spectral shape when Fig. 3(a) (*P. globosa*, $N = 18$, Chl-*/a* range = 7.3–45.6 mg/m^3) is compared with Fig. 3(b) (diatom, $N = 16$, Chl-*/a* range = 33.1–133.1 mg/m^3). The red (*P. globosa*) and blue (diatom) curves highlight the difference in R_{rs} when the two species showed similar Chl-*/a* concentrations (31.3 versus 33.1 mg/m^3), and this difference, mainly in the 400–550- and 575–625-nm spectral domains, was even more apparent in Fig. 3(c) after normalization at 570 nm. The second derivative of R_{rs} ($d\lambda^2 R_{rs}$) in the 450-490 nm spectral-domain further enhanced the subtle features associated with the two types of blooms [see Fig. 4(a)]. In agreement with findings reported by Lubac *et al.* [28], maxima close to ~475 nm were observed in the $d\lambda^2 R_{rs}$ spectra of *P. globosa* (red), while maxima close to ~460 nm were observed for diatoms (blue). Mean $d\lambda^2 R_{rs}$ spectra [dotted red and blue lines in Fig. 4(a)] indicated the same feature that there is a red-shift of the maxima in the $d\lambda^2 R_{rs}$ spectra for *P. globosa* compared to those of diatoms, which is consistent with the findings of Lubac *et al.* [28]. For all *in situ* R_{rs} collected in *P. globosa* bloom water in WTB and QZB ($N = 47$, Chl-*/a* ranged 2.4–45.6 mg/m^3), the maximum in the $d\lambda^2 R_{rs}$ spectra is found occurring in the range of 467.5–477.5 nm [see red circles in Fig. 4(b)]; on the other hand, it is in the range of 460.5–462.5 nm for diatom bloom water in the XMB (see blue circles in Fig. 4(b), $N = 16$, Chl-*/a* ranged 33.1–133.1 mg/m^3). It is noteworthy that the *P. globosa* bloom observations in WTB and QZB coincided with the formation of visible gelatinous colonies, suggesting that this species was dominant with a cell density of up to 10^6 cells/L [13].

Based on the results presented above, the approach of Lubac *et al.* [28] to differentiate *P. globosa* blooms from diatom blooms was followed by taking advantage of the shift of the maxima in the $d\lambda^2 R_{rs}$ spectra. After examining each $d\lambda^2 R_{rs}$ spectrum in the *in situ* data sets [see Fig. 4(b)], we defined a single empirical criterion for *P. globosa* identification, namely, the wavelength location of maximum $d\lambda^2 R_{rs}$ in the 460–480-nm range has to be longer than 466 nm.

B. Identification of P. globosa Using UAVs

The abovementioned method was then applied to the UAV-derived R_{rs} for the detection of *P. globosa*. In WTB, 54% of the UAV spots (after QC; see Section II-B) were identified as *P. globosa* (red circles), while diatoms (blue crosses) took up 31%, with the remaining 15% labeled as unknown, given that no local maximum of $d\lambda^2 R_{rs}$ in the 460–480-nm spectral domain was observed [see Fig. 5(a)]. In QZB, *P. globosa* and diatoms occupied 64% and 14%, respectively, and 22% remained unknown [see Fig. 5(b)]. The spots identified as *P. globosa* from UAV R_{rs} were, in general, also identified as *P. globosa* based on *in situ* R_{rs} (green triangles), where visible gelatinous colonies were indeed observed in the water column during samplings [see Fig. 5(a) and (b)] although most of these *in situ* stations were not concurrent with UAV flights. There were 16 match-ups of *in situ* and UAV-derived

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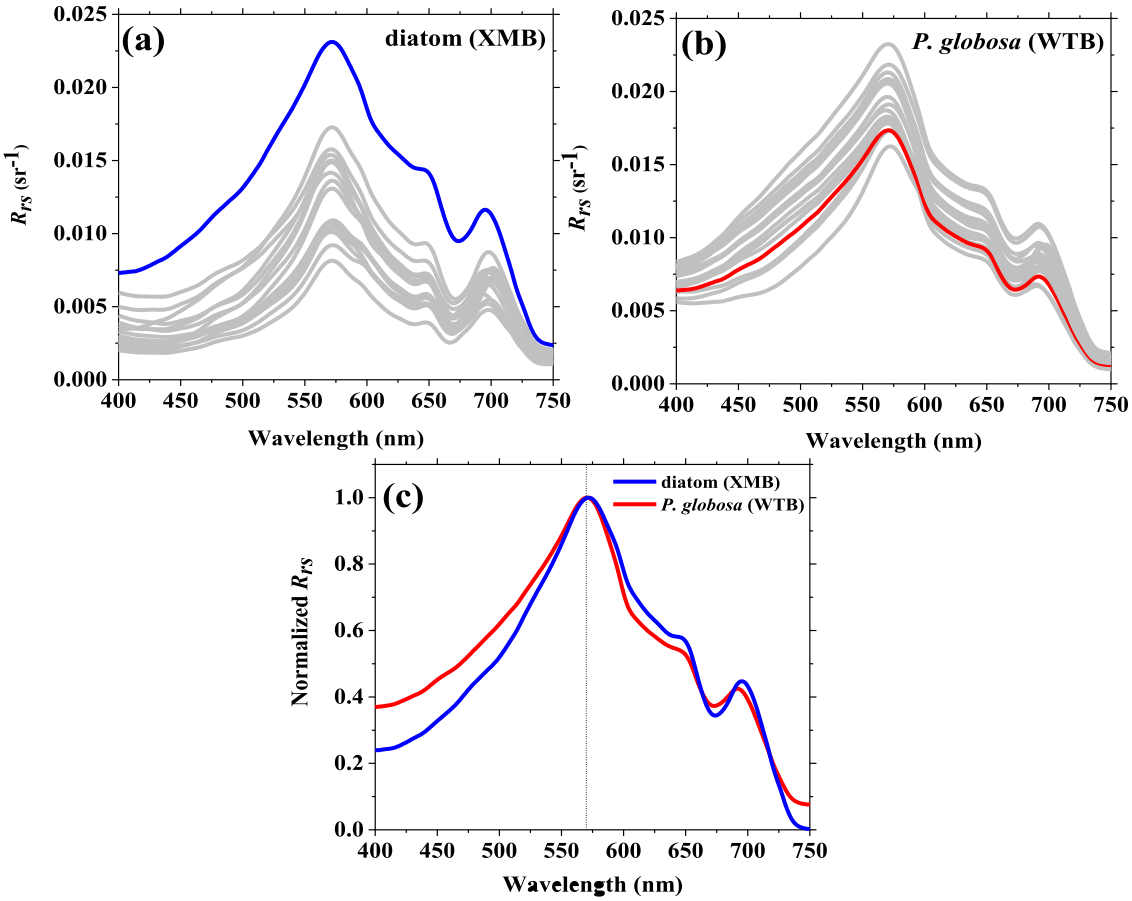


Fig. 3. (a) *In situ* measured remote sensing reflectance (R_{rs}) spectra of diatom blooms ($N = 16$) in XMB; the blue line highlights a sample with Chl-*/a* concentration of 33.1 mg/m^3. (b) *In situ* measured R_{rs} of *P. globosa* blooms ($N = 18$) in WTB; the red line highlights a sample with Chl-*/a* of 31.3 mg/m^3. (c) Comparison of the highlighted diatom (blue) and *P. globosa* (red) samples in (a) and (b) after normalization at 570 nm (see Section III-A for explanation).

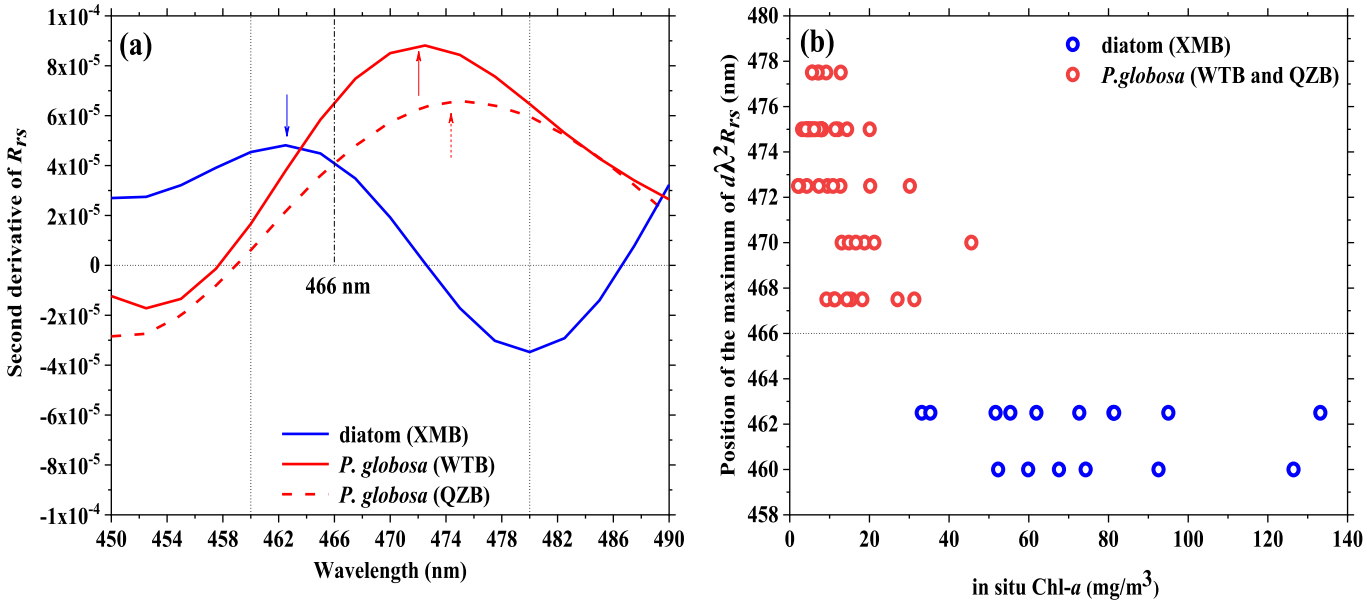


Fig. 4. (a) Selected second derivative of remote sensing reflectance ($d\lambda^2 R_{rs}$) spectra of the *P. globosa* bloom in WTB (solid red line for Chl-*/a* = 30.2 mg/m^3; red dashed line: Chl-*/a* = 12.1 mg/m^3) and diatom bloom in XMB (blue solid line: Chl-*/a* = 33.1 mg/m^3); the arrows are pointed to the position of $d\lambda^2 R_{rs}$ maxima. (b) Position of $d\lambda^2 R_{rs}$ maxima in 460–480 nm is derived from *in situ* measured remote sensing reflectance of *P. globosa* (red: $N = 47$) and diatom (blue: $N = 16$) blooms; their corresponding *in situ* measured Chl-*/a* was also shown for a better illustration.

R_{rs} within a window of 500 m and collected on the same day, as described in Section II-B. Twelve of these match-ups consistently resulted in the identification of *P. globosa*, while four of them identified as corresponding to diatoms based on the UAV-derived R_{rs} and the result from *in situ* R_{rs} pointed to *P. globosa* (in Table I, the four inconsistent match-ups

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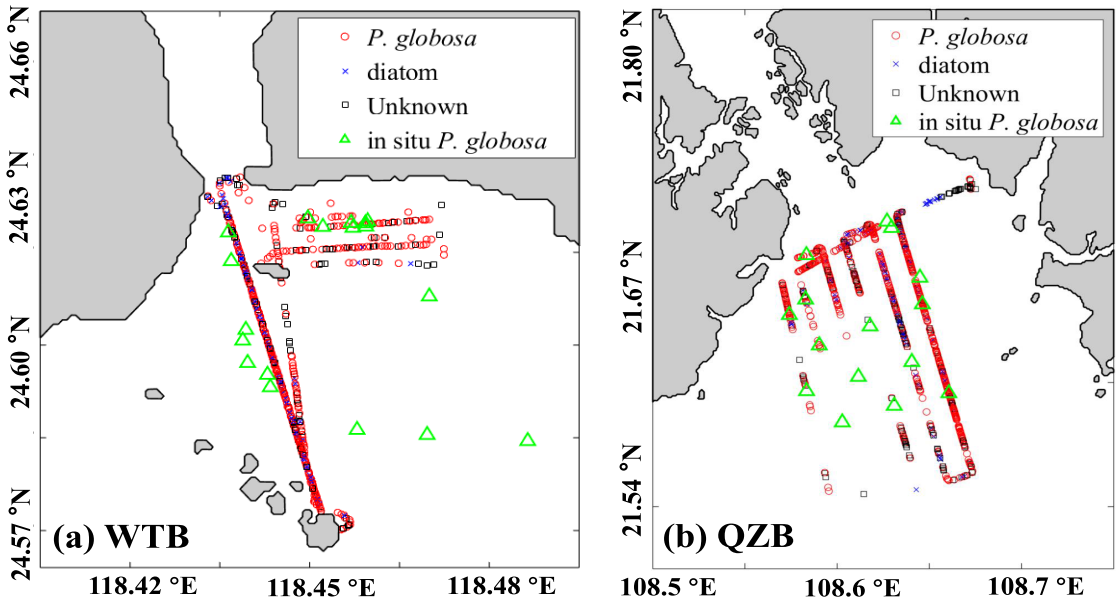


Fig. 5. Detection of *P. globosa* blooms (red circles) from remote sensing reflectance (R_{rs}) measured by UAV in (a) WTB and (b) QZB; some spots were identified as diatoms (blue crosses), and some have no maxima of $d\lambda^2 R_{rs}$ detected in 460–480 nm and, thus, labeled unknown (gray squares). Stations identified as *P. globosa* based on *in situ* measured remote sensing reflectance are shown for comparison (green triangles).

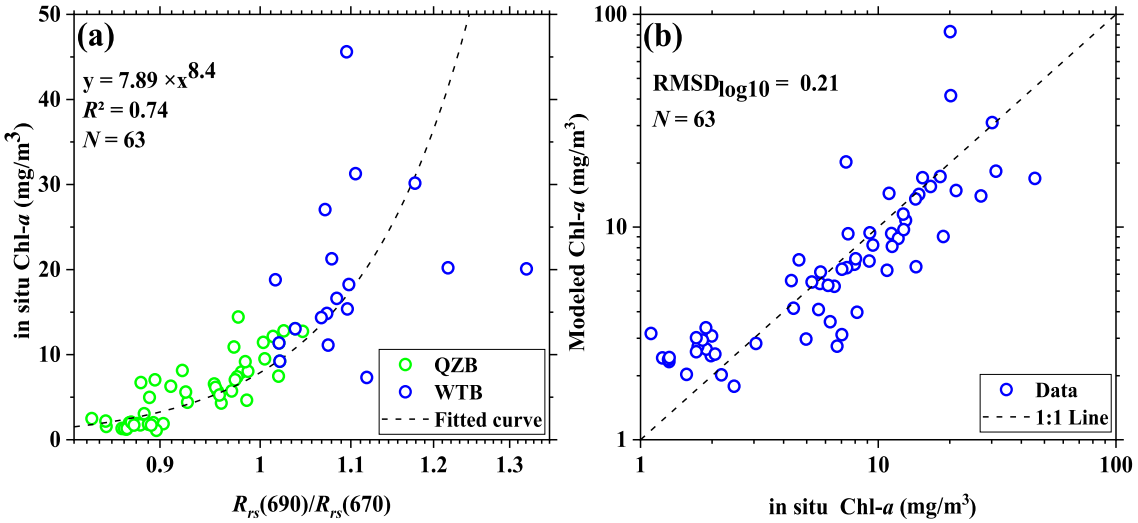


Fig. 6. (a) Relationship of remote sensing reflectance (R_{rs}) ratio at 690 and 670 nm versus Chl-*/a* in WTB (blue) and QZB (green), based on *in situ* measurements. (b) *In situ* Chl-*/a* versus modeled Chl-*/a*; R^2 = coefficient of determination; and $\text{RMSD}_{\log 10}$ = root-mean-square difference in the log10 scale.

are boldfaced). This result indicated an identification accuracy of 75%. Unfortunately, it is only known and confirmed that *P. globosa* was dominant in the water collected at limited sampling stations (see Fig. 1 for their location) after microscopic examination; no additional data were available to assess if those sites identified as dominated by diatoms were true assessments or not. However, the coexistence of *P. globosa* with diatoms is consistent with prior commonly reported observations [25], [47]–[49]. Indeed, in WTB and QZB, the dominant phytoplankton species is diatom under normal conditions [32]–[35].

C. Estimation of P. globosa Biomass Using UAVs

Following species identification, the magnitude of the *P. globosa* bloom was quantitatively assessed using Chl-*/a* concentration as the indicator. A number of empirical approaches to estimate Chl-*/a* from R_{rs} were tested and found that they did not perform well to the data set in this study [40], [50]–[52]. This failure of previous empirical models for Chl-*/a* estimation may be partially due to the unusual colony formation feature of *P. globosa*. The spherical colony is formed by small cells embedded in an organic mucilage, and the intracolonial space is fluid-filled. This may make the optical properties of pigments of this alga different from those of noncolony-forming algae. We, thus, conducted careful visual inspections of the R_{rs} spectrum and observed that the contrast of the troughs near the fluorescence peak may reflect Chl-*/a* changes. Based on match-ups of R_{rs} and Chl-*/a* collected *in situ*, an empirical relationship to estimate Chl-*/a* was developed by trial and error [coefficient of determination ($R^2 = 0.74$, $N = 63$, $P < 0.05$; see Fig. 6(a)].

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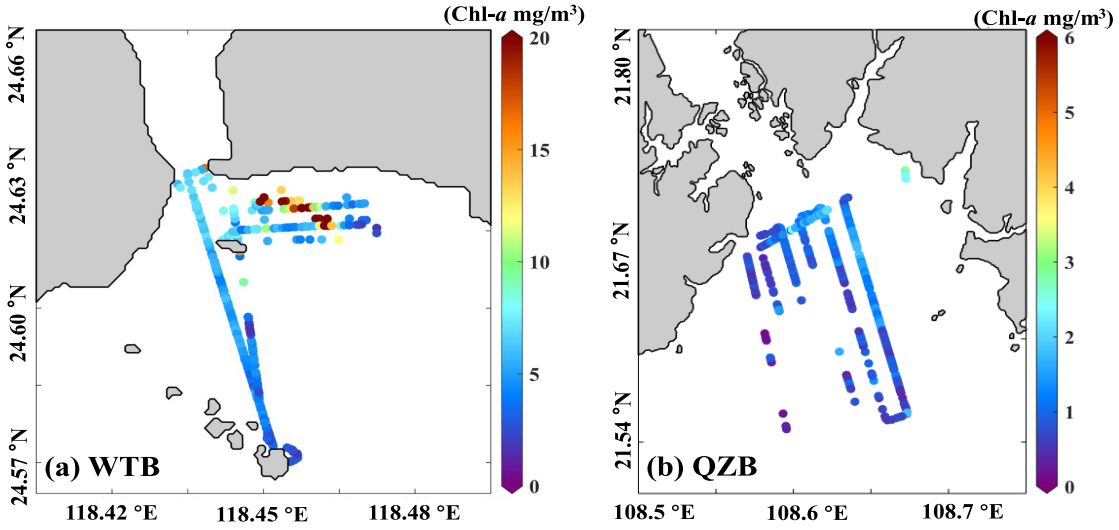


Fig. 7. Estimates of Chl-/a concentration from the UAV in (a) WTB and (b) QZB; those not identified as *P. globosa* (46% of the UAV spots in WTB and 36% of the UAV spots in QZB; see Section III-B) were excluded.

The resulted regression $\mathrm{RMSD}_{\log 10}$ (root-mean-square difference in log10 scale) is 0.21 [see Fig. 6(b)]. Due to the limited number of samples available, leave-one-out cross-validation (CV) was adopted to derive the above relationship. In the field of data mining and machine learning, leave-one-out CV has often been used for model evaluation and selection because it provides an almost unbiased estimation that minimizes manual intervention [53], [54]

$$\text{Chl-}/a = 7.8 \times \left[\frac{R_{\mathrm{rs}}(690)}{R_{\mathrm{rs}}(670)}\right]^{8.4}. \tag{4}$$

This localized empirical model [see (4)] specific to *P. globosa* was then used to estimate Chl-/a from UAV-derived R_{rs} at the spots identified as *P. globosa* (see Fig. 7). For the 16 matchups between UAV and *in situ* measurements, about half of the UAV-derived Chl-/a values were comparable to their corresponding *in situ* Chl-/a measurements (relative difference $< 50\%$), while the rest were underestimated by a factor of 2–4 (see Table I). The reasons leading to this discrepancy are discussed in Section IV-B.

Furthermore, an estimate of the number of colonies or cell counts is needed to better quantify the intensity of *P. globosa*. The Oceanic Environmental Monitoring and Forecasting Center of Qinzhou City collected 53 data groups including Chl-/a and colony counts of *P. globosa* in QZB during the period 2015–2018, with Chl-/a measured spectrophotometrically [55] and colonies counted visually. Based on this data set, an empirical relationship to estimate the number of colonies (Nc) from Chl-/a could be derived for a Chl-/a range of 1.5–306.6 mg/m^3 and a colony number range of 0–1600 gel matrix/L ($R^2 = 0.85$, $N = 53$, and $P < 0.05$; see Fig. 8)

$$N_c = 2.34 \times (\text{Chl-}/a)^{0.71}. \tag{5}$$

The number of *P. globosa* colonies in the two bays was then estimated from the UAV-derived Chl-/a estimates using (5). In WTB, higher numbers of colonies were observed in the northeastern part of the bay, with a maximum of up to 37 gel matrix/L [see Fig. 9(a)]. The value was not as high as the maximum of 200 gel matrix/L reported at the peak of the bloom by Fujian Marine Forecast [56]. Unfortunately, the UAV survey in this study was not carried out during the peak period of the bloom, and the *in situ* surveys with UAV overflights did not include colony enumeration. Meanwhile, estimates of colony count from UAV in QZB were ~3 gel matrix/L on average, i.e., an order of magnitude lower than those in WTB [see Fig. 9(b)]. This number agrees with visual observations during *in situ* samplings that yielded only a few colonies per liter of seawater. The colony number estimates at some stations, where UAV and *in situ* R_{rs} match-ups are located, are

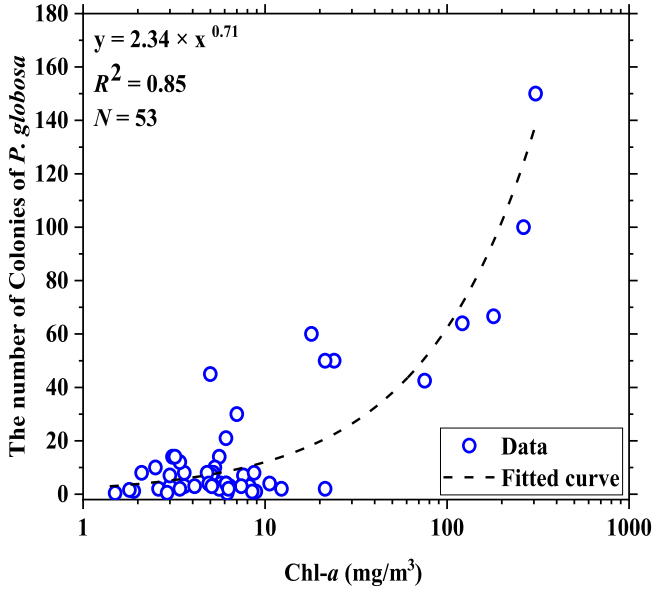


Fig. 8. Relationship of Chl-/a concentration versus *P. globosa* colony numbers in QZB based on a historical data set (see Section III-C); the fit power equation is indicated (R^2 as in Fig. 6).

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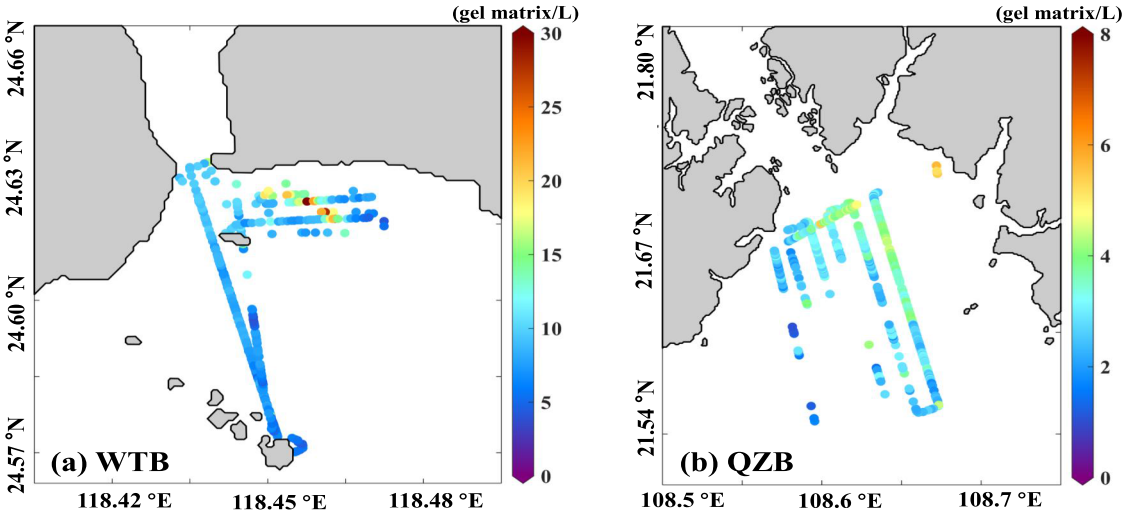


Fig. 9. Derived colony numbers of *P. globosa* from the UAV in (a) WTB and (b) QZB.

also listed in Table I. Results of those four stations identified as diatom dominant based on UAV R_{rs} are not listed.

IV. Discussion

A. Approach and Criteria for P. globosa Detection

As pointed out by Lubac *et al.* [28] and Astoreca *et al.* [25], what determines the separation of *P. globosa* from diatoms is mainly that there is a marker pigment for *P. globosa*, i.e., chlorophyll c_3 (Chl-/c_3), which is absent in diatoms. Although different strains of *P. globosa* may have different pigments [57] and the strains of *P. globosa* and diatom species in this study differ from those identified in the studies of Lubac *et al.* [28] and Astoreca *et al.* [25], similar diagnostic features were found, based on the measurements of algal culture described in Section II-C (see Fig. 10). Mean absorption (a_{ph}) spectra of cultured *P. globosa* and diatoms in this study were also similar to those reported by Astoreca *et al.* [25], showing a peak at ~466 nm for *P. globosa* [see Fig. 11(a)]. The absorption coefficient at ~466 nm [$a_{ph}(466)$] was found significantly different between diatoms and *P. globosa* (Mann–Whitney U-test, $P < 0.05$, and $N = 25$) and well correlated with the concentration of Chl-/c_3 in the *P. globosa* culture [see Fig. 11(b)]. It suggests that Chl-/c_3 present in *P. globosa* but missing in diatoms leads to a distinct absorption peak at ~466 nm prominent to *P. globosa*, thus leading to the successful separation of these two types of phytoplankton in the subtropical waters of this study. This is consistent with findings in temperate European coastal waters [25]. Therefore, the criterion at 466 nm for the identification of *P. globosa* based on $d\lambda^2 R_{rs}$, although somewhat empirical [see Fig. 4(b)], is rooted in the Chl-/c_3 absorption that is specific to *P. globosa* strains. This criterion is slightly different from that Lubac *et al.* [28] defined, which may be partially attributable to features associated with different *P. globosa* strains and diatom species, and/or different nonpigment optical properties in subtropical southern China coastal waters (this study) and those in temperate European coastal waters [25], [28].

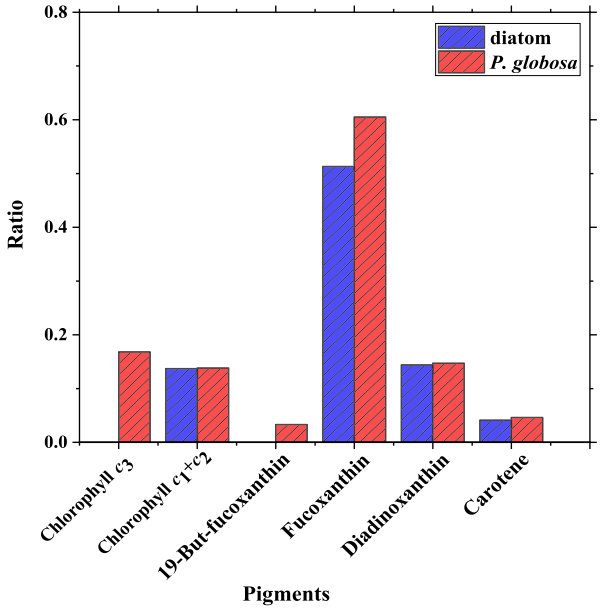


Fig. 10. Ratio of accessory pigments to Chl-/*a* in algal cultures, based on HPLC measurements. Blue: diatom. Red: *P. globosa*.

Also, it is important to determine whether there are any limitations of bloom conditions that would preclude the separation of the two types of phytoplankton. The WTB bloom [56] is characterized by a high-biomass bloom of *P. globosa* with up to 200 gel matrix/L, covering an area of ~150 km^2 at the peak. Colonies of *P. globosa* can often attain up to 2–3 cm in diameter according to field observations. In contrast, the QZB bloom is less intense, averaging ~3 gel matrix/L and ~0.2-0.7 cm in colony diameter. Therefore, at this stage, the QZB bloom can be conservatively viewed as the lower limit for *P. globosa* detection, i.e., at a density > 3 gel matrix/L and Chl-/a > 4 mg/m^3. It remains to be determined whether *P. globosa* blooms in their early stages,

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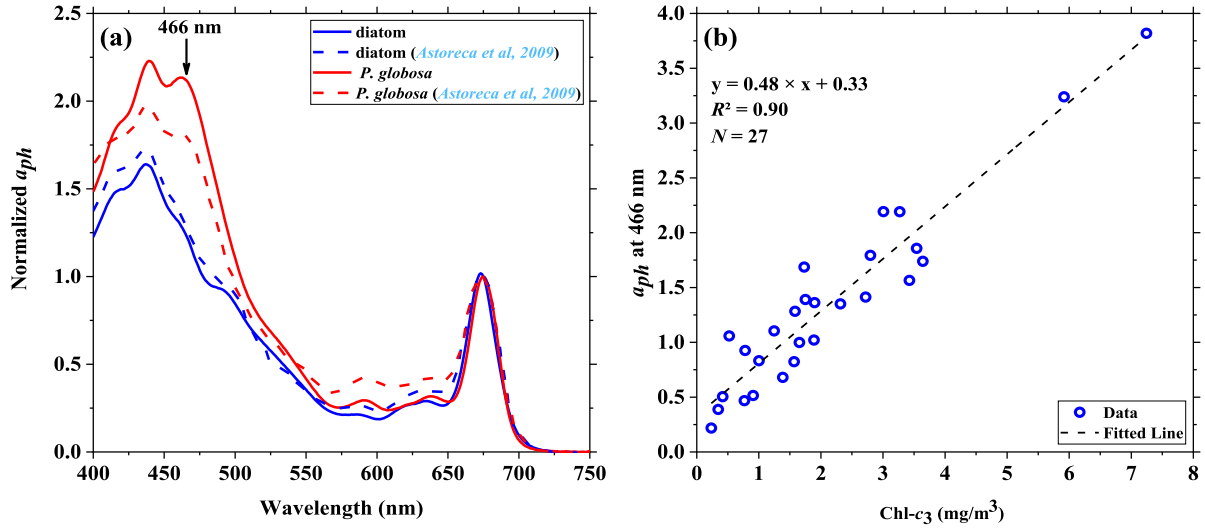


Fig. 11. (a) Absorption (a_{ph}) spectra of cultured *P. globosa* and diatoms normalized at 675 nm compared to those reported by Astoreca *et al.* [25]. (b) Relationship of measured Chl-/c_3 concentration and a_{ph} at 466 nm with *P. globosa* from the culture; the fit linear regression equation is indicated (R^2 as in Fig. 6).

e.g., before colony formation, can be identified using the criteria here proposed.

On the other hand, while it is well established that Chl-/c_3 absorption provides the basis for the discrimination of *P. globosa* from diatoms, we did not take the absorption-based algorithm developed by Astoreca *et al.* [25] for *P. globosa* identification. This is because there could be large uncertainties in deriving a_{ph} spectra from R_{rs} for such turbid coastal waters. For identification of *P. globosa* bloom, the approach based on the second derivative of R_{rs}, although empirical, is straightforward and easy to implement, which may also avoid some extra uncertainties from the derived a_{ph} spectrum.

B. Biomass Estimation of P. globosa

It is always desired to know the biomass of any phytoplankton, which is usually represented by the concentration of Chl-/*a*. However, it is found that the empirical model for Chl-/*a* estimate by Shang *et al.* [40], which was developed based on measurements during diatom and dinoflagellate blooms, was found not valid for Chl-/*a* of *P. globosa* blooms. Other models proposed to estimate Chl-/*a* in coastal waters [51], [58]–[60] were also tested in this study and found failed (results not shown). This failure is most likely attributed to the unusual feature of *P. globosa* in which it forms gelatinous colonies, as mentioned earlier. Thus, this study attempted to implement a novel empirical algorithm based on *in situ* measurements of R_{rs} and Chl-/*a* collected in *P. globosa* bloom waters, i.e., the WTB and QZB, and then applied it to UAV-derived R_{rs} to obtain Chl-/*a* estimates along the UAV flight tracks. The estimated Chl-/*a* from UAV-derived R_{rs} did not match perfectly with the *in situ* measured Chl-/*a* (see Table I). While it is possible that the model itself is experimental, this discrepancy could to some extent be induced by the uncertainty associated with the UAV-derived R_{rs} and the mismatch between the UAV spots and *in situ* sampling stations both in space and time. The latter may play an important role in the less satisfactory results in Table I because the study regions (WTB and QZB) are inner bays under the influence of semidiurnal tidal cycles. Thus, the Chl-/*a* value may vary a lot at any given site over hours although it could still be *P. globosa* bloom. In addition to the feature of the fast-changing environment, the distribution of phytoplankton is often not homogeneous.

The estimates of *P. globosa* colony number from Chl-/*a* concentration, therefore, have the same sources of uncertainty. Meanwhile, with the currently limited data, both from field surveys and culture experiments, it is not ready to determine the minimum *P. globosa* colonies or cell counts that could be detected with the current system. In addition, the UAV system used in this study has sensors for E_d and L_t only, which is not optimal for the correction of the surface reflected skylight. Better designed culture experiments and *in situ* surveys with a refined UAV system equipped with a sensor for L_{sky} would help to resolve these issues in the near future.

Nevertheless, the estimated results of colony numbers generated in this study appear reasonable compared to the very limited *in situ* observations, particularly in terms of the difference in the extent of the bloom between WTB and QZB. Most importantly, a rough estimate of colony numbers would be very helpful when a rapid response to HAB events is required by environmental agencies and the public. Nevertheless, it is important to note that models, such as those represented in (4) and (5), provide very localized empirical relationships. Caution should always be exerted in their application to other coastal waters.

V. Conclusion

This study takes the first attempt to detect and assess the extent and magnitude of *P. globosa* blooms in coastal waters using hyperspectral measurements obtained by UAV overflights. Encouraging results were obtained, with *P. globosa* successfully identified based on the second derivative of R_{rs}

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obtained from UAV-based radiometric measurements. Furthermore, for the first time, a distribution of *P. globosa* blooms (represented by colony number) at a spatial resolution of 30 m was obtained in WTB and QZB, China. The results also highlight the advantages of hyperspectral remote sensing for HAB monitoring in near-shore bays and estuarine waters [31], [61]. While more surveys of different types of phytoplankton blooms with UAV overflights are certainly required to reduce the uncertainties and to extend the coverage, the results here clearly demonstrated the feasibility of mapping blooms of *P. globosa* from UAV platforms. This is in particular important with the advancement of UAV technologies, where, at present, the conventional ocean color satellites can hardly meet the requirement of spectral, spatial, and temporal resolutions at the same time. As with all remote sensing techniques under development, there are always various limitations in its initial stage of a new approach or new system. We envision that this UAV-based scheme will be refined and improved after more and complete biological and optical data are collected.

ACKNOWLEDGMENT

The authors appreciate Prof. Bangqin Huang, Prof. Dazhi Wang, and Prof. Xin Liu of the State Key Laboratory of Marine Environmental Science (MEL), Xiamen University, for their help in high-pressure liquid chromatography (HPLC) measurements and algal culture. They also thank two anonymous reviewers for their constructive comments and suggestions that greatly improved this manuscript. This article is Contribution No. 2020396 of MEL.

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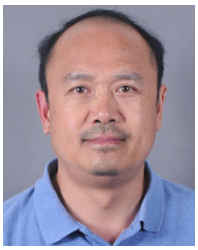
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